

Spin at the Closed Motion–Staticity Boundary: Rest-State Topology, Wigner Reduction, and Little-Group Contraction

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Abstract

For a fixed positive intrinsic scale Q , the positive-energy massive momentum orbit can be written in compact motion–staticity variables $D = p/\sqrt{Q^2 + |p|^2}$ and $S = Q/\sqrt{Q^2 + |p|^2}$, with $|D|^2 + S^2 = 1$ and $S > 0$. The resulting open ball has an isotropic rest point at $D = 0$ and a projective null boundary at $S = 0$. We use this closed state space to compare the massive spin- j carrier with its directional helicity decomposition and with the finite-helicity structures associated with null directions. The compact ball admits a useful spinorial description: $\rho(D) = \frac{1}{2}(\mathbf{1} + D \cdot \sigma)$ is positive for $S > 0$, satisfies $\det \rho = S^2/4$, and becomes a rank-one projector at the boundary. Thus the ideal boundary is naturally $\mathbb{CP}^1 \simeq S^2$.

For $D \neq 0$, the operator $n \cdot J$, $n = D/|D|$, splits the massive spin carrier into helicity eigenlines $L_h \rightarrow S^2$ with first Chern class $c_1(L_h) = -2h$. Every nonzero-helicity line is therefore obstructed from extending through the directionless rest state, while the complete spin multiplet is topologically trivial because the Chern classes cancel. A real blow-up of the rest point resolves the directional singularity. At the outer boundary a separate rank obstruction appears: a rank- $2j + 1$ massive carrier cannot become a rank-one fixed-helicity carrier by ordinary smooth restriction, so a boundary helicity sector requires an additional reduction.

The same compact parameter S controls both geometry and representation theory. Constant- S direction spheres have Gaussian curvature $K_S = S^2/(1 - S^2)$, while the appropriately normalized transverse stabilizer generators obey $[\Pi_1, \Pi_2] = iK_S J$. Hence the standard $\mathrm{SO}(3) \rightarrow \mathrm{E}(2)$ contraction at fixed null scale is accompanied by geometric flattening as $S \rightarrow 0$. The full massive Wigner–Thomas connection is the canonical $\mathrm{SU}(2)$ connection induced from $\mathrm{SL}(2, \mathbb{C}) \rightarrow \mathrm{SL}(2, \mathbb{C})/\mathrm{SU}(2)$; in a helicity-adapted frame its off-diagonal terms grow as S^{-1} . Projection to a fixed helicity line removes these terms and leaves the finite Berry connection with curvature $f_h = -h \, d\Omega_2$.

Finally, the extreme-helicity line provides a global converse to the pointwise rank reduction. With the holomorphic structure for which $L_{-j} \simeq \mathcal{O}(2j)$, Borel–Weil gives $H^0(\mathbb{CP}^1, L_{-j}) \simeq V_j$ up to the conventional dual action. Thus the $2j + 1$ -dimensional massive spin representation can be recovered from global sections of a rank-one boundary line. The construction organizes standard representation-theoretic, topological, and Wigner–Thomas structures on a common motion–staticity base; it does not identify the projective boundary with a finite-energy massless limit, nor does it derive a gauge field theory from the bundle reduction.

1 Introduction

Massive spin and massless helicity are usually introduced as distinct branches of Wigner’s classification of unitary representations of the Poincaré group [Wigner1939, Weinberg1995]. In four spacetime dimensions, a massive momentum has compact little group $\mathrm{SO}(3)$, or $\mathrm{SU}(2)$

after passing to the spin cover, and a spin- j particle carries a $(2j + 1)$ -dimensional internal space. A nonzero null momentum has little group $E(2)$; in the finite-helicity sector its translation subgroup acts trivially and the irreducible internal carrier is one dimensional. The two cases also differ globally. Nonzero-helicity massless states are naturally described by line bundles over the null-momentum directions, and the first Chern number is tied to helicity by $c_1 = -2h$ in the convention used below [PalmerducaQin2024, PalmerducaQin2025, Dragon2024].

The purpose of this paper is not to replace these standard results, but to place the massive and null directional structures on one compact base and to keep separate several operations that are often compressed into the phrase “massless limit.” For a fixed intrinsic scale $Q > 0$, define

$$D = \frac{p}{\sqrt{Q^2 + |p|^2}}, \quad S = \frac{Q}{\sqrt{Q^2 + |p|^2}}, \quad |D|^2 + S^2 = 1. \quad (1)$$

The massive orbit is the open ball $|D| < 1$. Its center $D = 0, S = 1$ is the unique rest momentum, while $S \rightarrow 0$ is the infinite-rapidity end. After forgetting the diverging radial scale, the outer boundary records only a future null direction. It is therefore a *projective* null boundary, not the finite-energy massless shell itself.

The compactification exposes two geometrically different degenerations. At the center, the momentum direction ceases to exist. Away from the center, however, $n = D/|D|$ is well defined and the massive spin- j carrier can be diagonalized with respect to $n \cdot J$. The resulting eigenlines are precisely the directional helicity bundles. Their nontrivial Chern classes show that a fixed nonzero-helicity line cannot be extended through the isotropic center, even though the complete massive spin carrier extends smoothly. At the outer boundary, by contrast, the direction remains well defined but the interpretation of the fiber changes: a rank- $2j + 1$ massive multiplet cannot become a rank-one fixed-helicity massless carrier by restriction alone. A reduction, quotient, constraint, or equivalent selection is necessary.

A second advantage of the compact variables is that the staticity parameter S also measures the approach to the little-group contraction. At fixed nonzero S the stabilizer is isomorphic to $SO(3)$; in an unscaled basis the transverse commutator is proportional to S^2 and becomes Abelian at $S = 0$. The constant- S direction spheres in the massive observer hyperboloid flatten at the same rate. This gives a direct geometric interpretation of the familiar Inönü–Wigner contraction [InonuWigner1953].

The spin transport has a parallel description. Wigner rotations and Thomas precession can be expressed as holonomy of a Berry-type connection on the massive momentum orbit [Oblak2018]. In the present variables the connection becomes singular in a helicity-adapted frame as $S \rightarrow 0$ because the terms that mix different helicities scale as S^{-1} . The helicity-preserving component stays finite. Thus the correct operation is not to take a naive matrix-valued boundary limit and then select a helicity, but to reduce to the desired helicity line before taking the projective null limit. We show that the massive Wigner connection is the canonical $SU(2)$ connection obtained by projecting the Maurer–Cartan form of $SL(2, \mathbb{C})$ along the symmetric-space fibration $SL(2, \mathbb{C}) \rightarrow SL(2, \mathbb{C})/SU(2)$. This identifies the bulk and boundary connections as two levels of the same homogeneous-space geometry.

The revision also adds a global complement to the rank obstruction. Pointwise, a one-dimensional boundary fiber cannot equal the restriction of a $(2j + 1)$ -dimensional massive fiber. Globally, however, the extreme-helicity line has degree $2j$ (with a suitable sign convention), and the Borel–Weil theorem realizes the entire spin- j representation as the space of holomorphic sections of that line. The result is standard representation theory [Woodhouse1992, Hall2015, Perelomov1986]; its role here is to show that local rank reduction need not mean loss of global spin information.

The paper is organized as follows. Section 2 develops the compact state space and its Hermitian/projective-spinor description. Section 3 analyzes the directional helicity splitting, the

rest-state obstruction, Chern cancellation, and the real blow-up. [Section 4](#) treats the boundary rank reduction and distinguishes null rays from fixed null momenta. [Section 5](#) relates staticity to the stabilizer contraction and to the geometry of constant- S spheres. [Section 6](#) identifies the Wigner connection with the canonical $\mathrm{SU}(2)$ connection and studies its helicity reduction. [Section 7](#) discusses curvature dilution and topological flux, while [Section 8](#) gives the Borel–Weil reconstruction and a short spin-cover parity remark. [Section 9](#) treats the finite-dimensional boundary criterion that excludes continuous-spin translations. The final section summarizes the resulting bulk–boundary picture and its limitations.

2 The normalized Lorentz–Hermitian cone and the closed MS ball

2.1 Compactification of the massive orbit

We use units $c = 1$ unless otherwise stated. Fix an intrinsic scale $Q > 0$ and write a positive-energy massive momentum as (ε, p) with $\varepsilon = \sqrt{Q^2 + |p|^2}$. The compact variables in [\(1\)](#) are $D = p/\varepsilon$ and $S = Q/\varepsilon$, while the inverse relations are $p = QD/S$ and $\varepsilon = Q/S$. Thus ordinary momentum space $\mathbb{R}_p^3$ is radially identified with the open unit ball B_D^3 . The center $D = 0$, $S = 1$ is the rest state, and the infinite-rapidity end $S \rightarrow 0$ becomes the boundary sphere $\partial B^3 = S^2$.

In rapidity coordinates, $|D| = \tanh \chi$ and $S = \operatorname{sech} \chi$. The massive observer space carries the hyperbolic metric

$$ds_{H^3}^2 = d\chi^2 + \sinh^2 \chi d\Omega_2^2. \quad (2)$$

Hence the open MS ball is the ball model of H^3 ; $S = 1$ at the unique rest point and $S \rightarrow 0$ at the ideal boundary.

2.2 The normalized Lorentz–Hermitian cone

A future Lorentz vector can be represented by a 2×2 Hermitian matrix. For the normalized MS state define

$$\rho(D) = \frac{1}{2}(\mathbf{1} + D \cdot \sigma), \quad (3)$$

where $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ are the Pauli matrices. The eigenvalues are $(1 \pm |D|)/2$, so $\rho(D)$ is positive definite for $|D| < 1$ and positive semidefinite on the closed ball. Its trace is one and

$$\det \rho(D) = \frac{1 - |D|^2}{4} = \frac{S^2}{4}. \quad (4)$$

Thus staticity is the determinant coordinate $S = 2\sqrt{\det \rho}$. The isotropic rest state is $\rho(0) = \mathbf{1}/2$, while the projective null boundary is the rank-one locus.

Proposition 2.1 (Normalized Lorentz–Hermitian cone). *The map $D \mapsto \rho(D)$ is an affine diffeomorphism from the closed motion ball $\overline{B}^3$ onto*

$$\mathcal{B}_{\text{MS}} = \{\rho = \rho^\dagger \geq 0 : \operatorname{Tr} \rho = 1\}.$$

Its interior consists of rank-two positive matrices and is naturally $H^3 \simeq \mathrm{SL}(2, \mathbb{C})/\mathrm{SU}(2)$. Its extreme points are the rank-one projectors, which form $\mathbb{CP}^1 \simeq S^2$.

Proof. Every trace-one Hermitian 2×2 matrix has a unique Pauli decomposition $\rho = \frac{1}{2}(\mathbf{1} + D \cdot \sigma)$. Positivity is equivalent to $1 \pm |D| \geq 0$, hence to $|D| \leq 1$. Rank one occurs exactly when $|D| = 1$, or equivalently $\det \rho = 0$. The rank-one positive trace-one matrices are precisely projectors onto complex lines in $\mathbb{C}^2$, giving $\mathbb{CP}^1$. The identification of the positive-definite determinant-one Hermitian matrices with $\mathrm{SL}(2, \mathbb{C})/\mathrm{SU}(2)$ is the standard Hermitian model of H^3 ; trace normalization gives the equivalent ball section used here. $\square$

The cone language makes two endpoint degenerations transparent. At $D = 0$ the two eigenvalues coincide, so no eigendirection is distinguished. At $S = 0$ one eigenvalue vanishes and the remaining eigenspace defines a complex line. Thus loss of direction at rest and emergence of a projective spinor line at the outer boundary are different rank phenomena.

2.3 Extreme points and convex decomposition

Let $D = dn$ with $0 \leq d \leq 1$ and $|n| = 1$, and define the antipodal rank-one projectors $\Pi_{\pm n} = \frac{1}{2}(\mathbf{1} \pm n \cdot \sigma)$. Then

$$\rho(dn) = \frac{1+d}{2} \Pi_n + \frac{1-d}{2} \Pi_{-n}. \quad (5)$$

The coefficients are the two eigenvalues of $\rho(dn)$. Consequently the closed MS ball is the convex hull of its projective-null extreme points. The rest state is the equal mixture $\rho(0) = \frac{1}{2}(\Pi_n + \Pi_{-n})$, independent of n , while the boundary limit $d \rightarrow 1$ leaves a single projector Π_n .

This convex statement is purely kinematical. In a gravitational application a total translation covector must first be shown to lie in the future Lorentz cone before it can be normalized to a point of $\mathcal{B}_{\text{MS}}$; that positivity question belongs to the companion analysis of null responses, not to the one-particle spin geometry developed here.

2.4 Projective spinors and null rays

Every extreme point can be written as

$$\Pi_n = \frac{\zeta \zeta^\dagger}{\zeta^\dagger \zeta} = \frac{1}{2}(\mathbf{1} + n \cdot \sigma), \quad |n| = 1, \quad (6)$$

for a nonzero two-spinor $\zeta \in \mathbb{C}^2$. Multiplication by any $c \in \mathbb{C}^\times$ leaves the projector unchanged, so the boundary is naturally

$$\partial \mathcal{B}_{\text{MS}} \simeq \mathbb{CP}^1 \simeq S^2. \quad (7)$$

Restricting to unit spinors gives the Hopf fibration $\text{U}(1) \rightarrow S^3 \rightarrow S^2$. The helicity line bundles studied below are associated line bundles over this projective boundary.

The boundary point specifies a future null *ray*, not a fixed null four-momentum. If $\ell = (1, n)$ is one representative, the projective boundary identifies $\ell \sim a\ell$ for $a > 0$. The stabilizer of the ray therefore contains the one-parameter boost that rescales the representative and is the Lorentz similitude subgroup $\text{Sim}(2)$, or the corresponding parabolic subgroup in the spin cover. Once a positive scale along the ray is fixed, one obtains a particular null momentum and the remaining little group is $\text{E}(2)$ [CohenGlashow2006, Weinberg1995]. This distinction is the group-theoretic version of the fact that $S \rightarrow 0$ at fixed Q is a projective infinite-rapidity limit rather than a finite-energy massless limit.

3 Directional helicity and the rest-state obstruction

3.1 Helicity splitting on the punctured ball

Let $V_j \simeq \mathbb{C}^{2j+1}$ be the irreducible spin- j representation of $\text{SU}(2)$, with Hermitian generators J_a satisfying $[J_a, J_b] = i\epsilon_{abc}J_c$. The complete massive spin carrier over the compact motion ball is the trivial bundle

$$\mathcal{E}_j = B^3 \times V_j. \quad (8)$$

Away from the rest state, define $n(D) = D/|D|$ and the directional helicity operator $H_n = n \cdot J$. Its simple spectrum is $h = -j, -j+1, \dots, j$, and the corresponding spectral projector may be

written as

$$\Pi_h(n) = \prod_{\substack{m=-j \\ m \neq h}}^j \frac{n \cdot J - m \mathbf{1}}{h - m}. \quad (9)$$

The image of $\Pi_h(n)$ is one dimensional. As n varies, these eigenspaces form a complex line bundle $L_h \rightarrow S^2$. Pulling L_h back along $n : B^3 \setminus \{0\} \rightarrow S^2$ gives a line bundle $\tilde{L}_h$ on the punctured motion ball.

Proposition 3.1 (Directional helicity splitting). *Over $B^3 \setminus \{0\}$ the massive spin carrier splits smoothly as*

$$\mathcal{E}_j|_{B^3 \setminus \{0\}} \simeq \bigoplus_{h=-j}^j \tilde{L}_h. \quad (10)$$

The splitting depends only on $n = D/|D|$ and is constant along each radial motion ray.

Proof. The projectors (9) depend smoothly on n and satisfy $\Pi_h \Pi_{h'} = \delta_{hh'} \Pi_h$ and $\sum_h \Pi_h = \mathbf{1}$. Their images therefore define a smooth direct-sum decomposition of the trivial carrier on the punctured ball. $\square$

The splitting exists at every nonzero momentum, including arbitrarily small momentum, but it has no canonical extension to $D = 0$ because n is undefined there. The next result shows that for $h \neq 0$ the failure is topological.

3.2 Chern class and nonextension through rest

For a helicity- h line over the direction sphere, the standard bundle result is

$$c_1(L_h) = -2h, \quad (11)$$

with the orientation and Berry-curvature convention used below [PalmerducaQin2025, Dragon2024]. The sign changes if both conventions are reversed, but nonvanishing for $h \neq 0$ does not.

Theorem 3.2 (Rest-state helicity obstruction). *If $h \neq 0$, the directional helicity line $\tilde{L}_h \rightarrow B^3 \setminus \{0\}$ does not extend as a complex line bundle over the full ball B^3 .*

Proof. The punctured ball deformation retracts to any sphere $S_r^2 = \{|D| = r\}$, $0 < r < 1$, and $\tilde{L}_h|_{S_r^2}$ has Chern number $-2h$. If an extension over B^3 existed, it would be topologically trivial because B^3 is contractible. Its restriction to S_r^2 would therefore have vanishing first Chern class, contradicting (11). $\square$

The obstruction is located at the isotropic rest point, not at the null boundary. The same line bundle occurs on every nonzero direction sphere, and its Chern number does not change with radius. In Berry-curvature language the topology is represented locally by a monopole-like two-form centered at the excluded point $D = 0$. This is a momentum-space bundle singularity, not a physical magnetic monopole.

3.3 The complete multiplet and Chern cancellation

The full massive carrier is unaffected by the obstruction because the helicity Chern classes cancel. On any direction sphere,

$$c_1\left(\bigoplus_{h=-j}^j L_h\right) = \sum_{h=-j}^j c_1(L_h) = -2 \sum_{h=-j}^j h = 0. \quad (12)$$

For spin one, for example, the three lines have Chern numbers $2, 0, -2$; for spin two they are $4, 2, 0, -2, -4$. The cancellation is consistent with the global triviality of $\mathcal{E}_j$.

This makes the role of the rest state precise. A complete spin multiplet can be carried through rest because no directional eigenline remains independently distinguished there. The obstruction appears only when one insists on retaining a nonzero-helicity line as a separate global subbundle.

3.4 Blowing up the rest state

A convenient resolution replaces the center of the ball by its sphere of approach directions. The real blow-up $[B^3; 0]$ is topologically $[0, 1] \times S^2$, with blow-down map $\beta : [B^3; 0] \rightarrow B^3$. The direction variable remains defined at the new inner boundary, so each helicity line pulls back to a smooth bundle on the resolved space.

Proposition 3.3 (Blow-up resolution). *Every helicity line extends smoothly over $[B^3; 0]$. For $h \neq 0$, the extended line cannot descend through β to a line bundle on B^3 .*

Proof. The resolved space is a product in the radial variable, so the pullback of $L_h \rightarrow S^2$ is smooth on the entire blow-up. If it descended to the ordinary ball, its restriction to any sphere surrounding the origin would come from a line bundle on a contractible space, contradicting [Theorem 3.2](#). $\square$

The blow-up therefore separates two descriptions that are conflated at the center of the ordinary ball: a globally smooth complete massive multiplet and a family of topologically nontrivial directional helicity lines.

4 Boundary rank reduction and the null fiber

The outer boundary presents a different problem. The complete massive spin- j fiber has complex dimension $2j + 1$, whereas a fixed finite-helicity massless carrier is one dimensional. A smooth vector bundle cannot change rank on a connected base.

Theorem 4.1 (Massive-to-helicity rank obstruction). *Let $j > 0$. There is no ordinary smooth complex vector bundle $\mathcal{V} \rightarrow \overline{B}^3$ whose restriction to the massive interior is a rank- $2j+1$ irreducible spin carrier and whose restriction to the boundary is a rank-one fixed-helicity line.*

Proof. The rank of a smooth vector bundle is locally constant and hence constant on the connected base $\overline{B}^3$. A bundle of rank $2j + 1$ on the interior therefore has the same rank at the boundary. $\square$

The theorem does not prescribe the mechanism of reduction. Depending on the field realization, the map from a complete massive multiplet to a fixed boundary helicity sector may involve a direct-sum splitting, a polarization constraint, a quotient, a gauge reduction, or a singular/stratified limit. The present statement is only that ordinary restriction is insufficient.

The natural intermediate object is the full directional boundary carrier

$$\mathcal{E}_j|_{S^2} = \bigoplus_{h=-j}^j L_h. \quad (13)$$

A fixed-helicity theory selects one summand. Schematically, if $\iota : S^2 \hookrightarrow \overline{B}^3$ is the boundary inclusion, the operation is $\mathcal{E}_j \rightarrow \iota^* \mathcal{E}_j \rightarrow L_h$. The first arrow concerns the base geometry; the second concerns the internal carrier. The distinction will be essential for the Wigner connection,

whose full boundary limit is singular even though the induced connection on a fixed helicity line is finite.

The projective nature of the boundary should also be kept separate from the physical massless orbit. The boundary point specifies a null ray $[\ell]$, so its stabilizer is the ray stabilizer $\text{Sim}(2)$ (or the corresponding parabolic subgroup in the spin cover). After a positive scale along the ray is fixed, one obtains a particular null momentum and the scale-preserving little group $E(2)$. Thus the base compactification, little-group contraction, and fiber reduction are three related but distinct operations.

5 Staticity and little-group contraction

5.1 Staticity-dependent stabilizer algebra

Choose a momentum along the third axis, $P = (\varepsilon, 0, 0, p)$, and set $\beta = p/\varepsilon$ and $S = Q/\varepsilon = \sqrt{1 - \beta^2}$. Let J_i and K_i be rotation and boost generators with $[J_i, J_j] = i\epsilon_{ijk}J_k$, $[J_i, K_j] = i\epsilon_{ijk}K_k$, and $[K_i, K_j] = -i\epsilon_{ijk}J_k$. The infinitesimal stabilizer of P is generated by

$$J = J_3, \quad T_1 = J_1 + \beta K_2, \quad T_2 = J_2 - \beta K_1. \quad (14)$$

A direct calculation gives

$$[J, T_1] = iT_2, \quad [J, T_2] = -iT_1, \quad [T_1, T_2] = iS^2 J. \quad (15)$$

For every $S > 0$, the rescaled generators $\hat{T}_a = T_a/S$ satisfy the ordinary $\mathfrak{so}(3)$ algebra. At $S = 0$, the unscaled T_1, T_2 commute and the algebra becomes $\mathfrak{e}(2)$. This is the standard group contraction, now parameterized by the motion-staticity scalar.

The rescaling T_a/S is singular at the boundary. That singularity is not a defect: it is precisely how a compact stabilizer becomes noncompact in an Inönü–Wigner contraction [InonuWigner1953]. The unscaled basis is the one that stays finite and turns the two transverse rotations into translations.

5.2 Constant-staticity geometry

From (2) and $S = \text{sech } \chi$, a surface of fixed staticity has induced metric

$$q_S = \frac{1 - S^2}{S^2} d\Omega_2^2. \quad (16)$$

Its intrinsic radius and Gaussian curvature are

$$R_S = \frac{\sqrt{1 - S^2}}{S}, \quad K_S = \frac{S^2}{1 - S^2}. \quad (17)$$

To compare the geometry directly with the stabilizer algebra, define $\Pi_a = T_a/\sqrt{1 - S^2}$ for $0 < S < 1$. Then

$$[J, \Pi_1] = i\Pi_2, \quad [J, \Pi_2] = -i\Pi_1, \quad [\Pi_1, \Pi_2] = iK_S J. \quad (18)$$

Thus $S \rightarrow 0$ is simultaneously the flattening limit $K_S \rightarrow 0$ and the contraction $\mathfrak{so}(3) \rightarrow \mathfrak{e}(2)$. At the opposite end, $S \rightarrow 1$ collapses the direction sphere to the rest point, with $R_S \rightarrow 0$ and $K_S \rightarrow \infty$. The two endpoints therefore encode different degenerations: loss of direction at rest and local flattening at the projective null edge.

The area form is $dA_S = (1 - S^2)S^{-2}d\Omega_2$, so $K_S dA_S = d\Omega_2$ and $\int K_S dA_S = 4\pi$ for every $0 < S < 1$. This compensation between vanishing pointwise curvature and diverging area will reappear in the helicity Chern flux.

5.3 Pauli–Lubanski invariant

With the same momentum convention, the Pauli–Lubanski components may be written, up to an overall sign convention, as $W^0 = pJ$, $W^1 = \varepsilon T_1$, $W^2 = \varepsilon T_2$, and $W^3 = \varepsilon J$. Hence

$$-\frac{W^2}{\varepsilon^2} = T_1^2 + T_2^2 + S^2 J^2. \quad (19)$$

For a massive spin- j irrep, $-W^2 = Q^2 j(j+1)$ and $Q = \varepsilon S$, so

$$T_1^2 + T_2^2 + S^2 J^2 = S^2 j(j+1). \quad (20)$$

At fixed finite j , the transverse contribution therefore vanishes as $S \rightarrow 0$. A continuous-spin limit would require a correlated scaling in which the spin label grows as the invariant mass scale is reduced, so that the transverse $E(2)$ translation invariant remains nonzero. That regime is not the fixed- j , fixed- Q projective compactification considered here.

6 Wigner transport as the canonical bulk connection

6.1 Canonical connection on $\mathrm{SL}(2, \mathbb{C})/\mathrm{SU}(2)$

The future unit timelike hyperboloid is the Riemannian symmetric space $H^3 \simeq \mathrm{SL}(2, \mathbb{C})/\mathrm{SU}(2)$. A convenient positive Hermitian representative of the standard boost from rest to rapidity χ in direction n is

$$g(\chi, n) = \cosh \frac{\chi}{2} \mathbf{1} + \sinh \frac{\chi}{2} n \cdot \sigma. \quad (21)$$

Write the Cartan decomposition of the Lie algebra as $\mathfrak{sl}(2, \mathbb{C}) = \mathfrak{su}(2) \oplus \mathfrak{p}$. The canonical principal connection on $\mathrm{SL}(2, \mathbb{C}) \rightarrow H^3$ is the $\mathfrak{su}(2)$ part of the Maurer–Cartan form, $A_K = \mathrm{pr}_{\mathfrak{su}(2)}(g^{-1}dg)$.

At fixed χ , with $c = \cosh(\chi/2)$ and $s = \sinh(\chi/2)$, one has $g^{-1} = c\mathbf{1} - s n \cdot \sigma$ and $dg = s dn \cdot \sigma$. Since $n \cdot dn = 0$ and $(n \cdot \sigma)(dn \cdot \sigma) = i(n \times dn) \cdot \sigma$, the compact part is

$$A_K = -is^2(n \times dn) \cdot \sigma = -i(\gamma - 1)(n \times dn) \cdot J \quad (22)$$

for spin one-half, where $J = \sigma/2$ and $\gamma = \cosh \chi = 1/S$. Applying the spin- j representation gives the same formula with the corresponding matrices J_a .

Proposition 6.1 (Canonical origin of the Wigner connection). *For the standard rotation-free boost section (21), the Wigner–Thomas connection on the massive spin- j carrier is the connection induced from $A_K = \mathrm{pr}_{\mathfrak{su}(2)}(g^{-1}dg)$. On angular directions at fixed staticity,*

$$A_W = -i(\gamma - 1)(n \times dn) \cdot J, \quad (23)$$

which is equivalent to $-i\gamma^2(\gamma + 1)^{-1}(D \times dD) \cdot J$.

This identifies the Wigner connection as a standard homogeneous-space object rather than an additional structure attached by hand to the compact variables. Its holonomy gives the usual Wigner rotation/Thomas precession [Oblak2018].

6.2 Helicity-adapted frame and the singular unreduced limit

Write $n = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ and choose

$$U(\theta, \phi) = e^{-i\phi J_3} e^{-i\theta J_2}, \quad U J_3 U^{-1} = n \cdot J. \quad (24)$$

In this frame the helicity eigenstates are the fixed J_3 eigenvectors. The transformed connection is $A_W^{\text{hel}} = U^{-1}A_W U + U^{-1}dU$, which evaluates to

$$A_W^{\text{hel}} = -i(\gamma J_2 d\theta - \gamma J_1 \sin \theta d\phi + J_3 \cos \theta d\phi). \quad (25)$$

The first two terms change the J_3 weight and grow as $\gamma = S^{-1}$. The diagonal term preserves helicity and remains finite.

Proposition 6.2 (Singularity of the unreduced spin connection). *For every nontrivial finite spin representation $j > 0$, the full Wigner connection in a helicity-adapted frame contains off-diagonal matrix elements of order S^{-1} . Consequently, the unreduced matrix-valued connection has no finite naive limit at $S = 0$.*

The divergence is a mixing effect between different helicity weights. It does not imply a divergence of the connection induced on an individual helicity line.

6.3 Reduction before the boundary limit

Let $\Pi_h(n)$ be the projector onto the helicity- h line. The induced covariant derivative is $\nabla_h = \Pi_h(d + A_W)\Pi_h$. In the helicity frame, $\langle h|J_1|h\rangle = \langle h|J_2|h\rangle = 0$ and $J_3|h\rangle = h|h\rangle$, so

$$A_h = -ih \cos \theta d\phi, \quad a_h = h \cos \theta d\phi, \quad f_h = da_h = -h \sin \theta d\theta \wedge d\phi. \quad (26)$$

Here $A_h = -ia_h$ is the anti-Hermitian line connection and f_h its real curvature.

Theorem 6.3 (Helicity reduction before the null-boundary limit). *The full Wigner connection develops helicity-mixing terms of order S^{-1} as $S \rightarrow 0$, whereas the induced connection on a fixed helicity line is independent of S and has curvature (26). Hence the finite boundary connection is obtained by reducing to a helicity eigenspace before taking the projective null limit.*

The local one-form a_h is not globally smooth for $h \neq 0$. On northern and southern patches one may use $a_h^{(N)} = h(\cos \theta - 1)d\phi$ and $a_h^{(S)} = h(\cos \theta + 1)d\phi$. Their difference is $-2h d\phi$, so the transition function has winding number $-2h$, reproducing (11).

7 Curvature dilution and fixed Chern flux

The relation between staticity and the direction-sphere geometry gives a compact form for the helicity curvature. From (17), $K_S dA_S = d\Omega_2$, and therefore

$$f_h = -h K_S dA_S. \quad (27)$$

As $S \rightarrow 0$, the pointwise norm of the curvature with respect to q_S tends to zero because $|f_h|_{q_S} = |h|K_S$, while the area of the direction sphere diverges like S^{-2} . The product remains fixed:

$$\int_{\Sigma_S} f_h = -4\pi h, \quad c_1(L_h) = \frac{1}{2\pi} \int_{\Sigma_S} f_h = -2h. \quad (28)$$

Thus local flattening of the constant-staticity spheres is compatible with a nonzero global topological charge. The Chern flux is diluted over an intrinsic sphere whose radius diverges.

The opposite endpoint gives the complementary picture. As $S \rightarrow 1$, the direction sphere collapses to the isotropic center while its Gaussian curvature diverges. A nonzero Chern flux cannot be pushed smoothly through that collapse while keeping a single helicity line distinguished. This is the differential-geometric counterpart of the rest-state obstruction in Theorem 3.2.

8 Global reconstruction from the extreme-helicity line

The rank obstruction in [Theorem 4.1](#) is pointwise: a one-dimensional fiber cannot equal a $(2j + 1)$ -dimensional fiber. It does not imply that the boundary line loses the global information of the spin- j representation. For the extreme helicity line there is a standard converse supplied by Borel–Weil.

Equip $\mathbb{CP}^1$ with its standard complex structure. In the sign convention $c_1(L_h) = -2h$, the negative extreme-helicity line has degree $2j$ and can be identified, after choosing the compatible holomorphic structure, with $\mathcal{O}(2j)$. The Borel–Weil theorem for $SU(2)$ then gives the spin- j irreducible representation as the space of holomorphic sections of this line, up to the conventional choice of dual action [[Woodhouse1992](#), [Hall2015](#), [Perelomov1986](#)].

Theorem 8.1 (Extreme-line reconstruction). *Let $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ and choose conventions such that $L_{-j} \simeq \mathcal{O}(2j)$. Then*

$$H^0(\mathbb{CP}^1, L_{-j}) \simeq V_j \quad (29)$$

up to dualization determined by the chosen left/right group action. In particular,

$$\dim H^0(\mathbb{CP}^1, L_{-j}) = 2j + 1 = c_1(L_{-j}) + 1. \quad (30)$$

Proof. For $k \geq 0$, the holomorphic sections of $\mathcal{O}(k)$ are homogeneous polynomials of degree k in two complex variables, a vector space of dimension $k + 1$. The natural $SU(2)$ action on these polynomials is the symmetric-power representation $\text{Sym}^k(\mathbb{C}^2)$, which is irreducible of spin $k/2$. Taking $k = 2j$ gives the result. $\square$

This theorem sharpens the interpretation of the boundary reduction. The fixed helicity fiber is one dimensional, but the global section space of the extreme line has the correct $(2j + 1)$ -dimensional multiplicity. Local rank reduction and global representation content are therefore different questions. The full massive multiplet is not the same bundle as L_{-j} , but its representation can be reconstructed from the global holomorphic geometry of that boundary line.

The same observation also links the Chern class to the spin-cover parity. Any helicity weight h in a spin- j multiplet satisfies $j - h \in \mathbb{Z}$, hence

$$(-1)^{2j} = (-1)^{2h} = (-1)^{c_1(L_h)}. \quad (31)$$

This is only a mod-two shadow of the full Chern class: integer-spin helicity lines can be topologically nontrivial even though the 2π spin-cover parity is trivial.

9 Finite-dimensional boundary fibers and continuous spin

After a null scale has been chosen, the massless little group is $E(2) = SO(2) \ltimes \mathbb{R}^2$ (or a corresponding cover). Besides ordinary helicity representations, it admits continuous-spin representations in which the translation subgroup acts nontrivially. The compact boundary construction by itself does not rule out that branch. A simple finite-dimensional criterion does.

Theorem 9.1 (Finite-dimensional boundary criterion). *Let U be a continuous finite-dimensional unitary irreducible representation of $E(2)$, or of the cover appropriate to the chosen spin structure. Then the translation subgroup acts trivially. The representation therefore reduces to a one-dimensional character of the rotational subgroup.*

Proof. Let P_1, P_2 be the commuting Hermitian generators of the translation subgroup. Their joint spectrum is a finite subset $\Sigma \subset \mathbb{R}^2$. Rotational covariance requires Σ to be invariant under

all planar rotations. A nonzero point would generate an infinite circular orbit, so $\Sigma = \{0\}$. Hence $P_1 = P_2 = 0$. The representation then factors through an Abelian rotational group, whose finite-dimensional unitary irreducible representations are one dimensional. $\square$

The quantization of the helicity label depends on the global cover. In the fixed finite- j contraction of (20), the transverse translation invariant vanishes automatically as $S \rightarrow 0$, consistent with the finite-helicity branch. A genuine continuous-spin limit requires a different correlated scaling, roughly $j \sim S^{-1}$, so that the Pauli–Lubanski scale remains finite while the massive invariant scale goes to zero.

10 Discussion and conclusion

The closed motion–staticity ball provides a common base for several structures that are usually discussed separately. Its interior is the massive observer space $H^3 \simeq \mathrm{SL}(2, \mathbb{C})/\mathrm{SU}(2)$, written in compact variables. Its boundary is the projective null sphere $\mathbb{CP}^1$, equivalently the rank-one locus of the normalized positive Hermitian cone. Staticity is the determinant coordinate $S = 2\sqrt{\det \rho}$ and controls the approach from rank two to rank one.

The rest point and the null boundary play sharply different roles. At rest, directional helicity ceases to be defined. A nonzero-helicity eigenline cannot be extended through the center because $c_1(L_h) = -2h$, while the full multiplet extends because the Chern classes cancel. At the outer boundary, the direction remains meaningful but the internal interpretation changes. A rank- $2j+1$ carrier does not become a rank-one helicity carrier by restriction; a separate reduction is required. The full boundary restriction retains all directional weights simultaneously.

The homogeneous-space description unifies the transport picture. The standard Wigner–Thomas connection is the canonical $\mathrm{SU}(2)$ connection on $\mathrm{SL}(2, \mathbb{C}) \rightarrow \mathrm{SL}(2, \mathbb{C})/\mathrm{SU}(2)$. In a helicity frame, the transverse generators mix weights with coefficients of order S^{-1} , so the unreduced connection has no finite boundary matrix limit. Projection to a $\mathrm{U}(1)$ weight line removes the mixing and leaves the Berry connection whose Chern flux is fixed. The same parameter that contracts the little algebra flattens the constant-staticity direction spheres, while the topological flux survives because the intrinsic area diverges.

The Borel–Weil reconstruction gives a useful complement to the rank obstruction. Although a boundary helicity fiber is one dimensional, the extreme-helicity line of degree $2j$ has a $(2j+1)$ -dimensional holomorphic section space carrying the spin- j irrep. The massive multiplicity can therefore be encoded globally in boundary line-bundle geometry even though it is not preserved pointwise as fiber rank.

Several limits should not be conflated. The boundary $S = 0$ is a space of null rays. It does not by itself supply the finite energy scale of a massless momentum. Consequently the stabilizer of the projective direction is larger than the scale-fixed massless little group; the $\mathrm{E}(2)$ algebra appears after a representative null momentum is chosen. Likewise, the line-bundle reduction does not derive gauge symmetry or a covariant massless field theory. Longitudinal decoupling in vector or tensor fields, interacting amplitudes, and local gauge constraints require additional field-theoretic structure.

There is also a natural relation to the asymptotic response program, but it belongs to a separate level of analysis. A projective null direction on the same $\mathbb{CP}^1$ can be used to test a Lorentzian boundary charge through directional responses, and a spinorial lift of that direction enters Witten-type positive-energy constructions. Those questions concern gravitational constraints and boundary Hamiltonians rather than one-particle spin transport; they are not needed for the present results. Keeping the levels separate is useful: the present paper concerns the geometry of the compact state base and its spin/helicity fibers, while positive-energy and quantum-boundary constructions require additional dynamics.

The resulting hierarchy can be summarized as follows: the massive base is B^3 ; away from rest the spin carrier splits into helicity lines; the nonzero Chern class obstructs extension of an individual line through the center; the outer limit contracts the scale-fixed stabilizer from $\text{SO}(3)$ to $\text{E}(2)$; the full Wigner connection must be reduced before the projective null limit; and the extreme-helicity line globally reconstructs the massive spin representation through Borel–Weil. These statements organize standard Lorentz representation theory, bundle topology, and Wigner–Thomas transport on one closed motion–staticity geometry.

A Pauli–Lubanski conventions

For completeness, take metric signature $(+---)$ and momentum $P^\mu = (\varepsilon, 0, 0, p)$. With $W^0 = \mathbf{P} \cdot \mathbf{J}$ and $\mathbf{W} = P^0 \mathbf{J} - \mathbf{P} \times \mathbf{K}$, one has

$$W^0 = pJ_3, \quad W^1 = \varepsilon J_1 + pK_2 = \varepsilon T_1, \quad W^2 = \varepsilon J_2 - pK_1 = \varepsilon T_2, \quad W^3 = \varepsilon J_3. \quad (32)$$

Consequently

$$W_\mu W^\mu = -\varepsilon^2 (T_1^2 + T_2^2 + S^2 J^2), \quad (33)$$

where $p^2 = \varepsilon^2(1 - S^2)$. For a massive spin- j irrep, $W^2 = -Q^2 j(j+1)$ and $Q = \varepsilon S$, which gives (20). The overall sign of W^μ depends on the convention for $\epsilon^{\mu\nu\rho\sigma}$ and does not affect the contraction argument.

B Projector curvature and the helicity Chern number

Let $|h\rangle$ be a normalized eigenvector of J_3 with eigenvalue h and define $|h; n\rangle = U(\theta, \phi)|h\rangle$ using (24). The rank-one projector is $\Pi_h(n) = |h; n\rangle\langle h; n|$. A local real Berry potential is

$$a_h = i\langle h; n | d | h; n \rangle = h \cos \theta d\phi, \quad (34)$$

so $f_h = da_h = -h \sin \theta d\theta \wedge d\phi = -h d\Omega_2$. Equivalently, the curvature can be written in projector form as

$$f_h = i \text{Tr}(\Pi_h d\Pi_h \wedge d\Pi_h). \quad (35)$$

It follows that

$$c_1(L_h) = \frac{1}{2\pi} \int_{S^2} f_h = -2h. \quad (36)$$

After radial pullback to $B^3 \setminus \{0\}$, the curvature has no radial component. In Cartesian motion coordinates it takes the monopole form

$$f_h = -\frac{h}{2} \epsilon_{abc} \frac{D^a}{|D|^3} dD^b \wedge dD^c, \quad (37)$$

whose flux through every sphere surrounding the origin is $-4\pi h$. The singularity at $D = 0$ expresses the failure of the directional eigenline to descend through the isotropic rest state; on the real blow-up it is a regular two-form on each direction sphere.